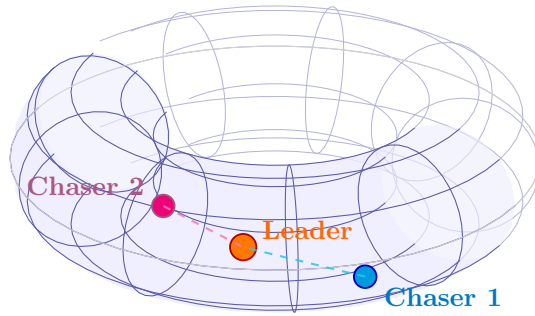


Stochastic Pursuit on a Dynamic Torus

A Multi-Agent Delay-Pursuit System with Metric Perturbations,
Rendered via Vulkan

Research & Study Notes · Solo Independent Study

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$\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ with mesh, leader (●) and chasers (● ●). Dashed lines indicate pursuit geodesics.

Mission Statement

This project is a long-horizon study in the unity of pure mathematics and high-performance computation. The goal is to build, from first principles, a real-time simulation of **multiple-agent Brownian motion on a two-dimensional torus** $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, in which some agents pursue others with a *time-delayed* information signal, and in which the geometry of the arena is itself a dynamic, user-perturbed object.

The simulation is not an end. It is a *forcing function*—a concrete artifact that demands genuine understanding of every layer of mathematics beneath it. A renderer that produces a wrong geodesic is a proof of misunderstanding. A pursuer that behaves incorrectly under a metric ripple is a failed theorem. The Vulkan pipeline is, in this sense, an automated grader with no partial credit.

The work proceeds without shortcuts and without a deadline. Each stage of the

dependency graph below must be structurally understood before the next is approached. This document is the living front matter of that journey.

Problem Statement

The Arena

Let $\mathbb{T}^2$ denote the flat two-dimensional torus, realized as the quotient:

$$\mathbb{T}^2 \cong \mathbb{R}^2 / \mathbb{Z}^2,$$

equipped with the background Riemannian metric $g_0 = dx^2 + dy^2$. This metric is flat (zero Gaussian curvature) and induces a geodesic distance

$$d_0(x, y) = \min_{(m, n) \in \mathbb{Z}^2} \|(x - y) + (m, n)\|_{\mathbb{R}^2}.$$

The arena is compact, boundary-free, and doubly periodic—a natural domain for stochastic processes that must neither escape to infinity nor reflect from walls.

The Dynamic Metric

At any time $t \geq 0$ and in response to user input (a “poke” at point $p \in \mathbb{T}^2$ initiated at time t_0), the metric is deformed conformally:

$$g(x, t) = (1 + \varepsilon h(x, t; p, t_0)) g_0, \quad \varepsilon A < 1,$$

where h is a smooth, spatially propagating, temporally decaying perturbation field. The positive-definiteness condition $\varepsilon A < 1$ guarantees that $g(x, t)$ remains a valid Riemannian metric for all time. The perturbed geodesic distance satisfies, to first order in ε :

$$d_g(x, y, t) = d_0(x, y) + \frac{\varepsilon}{2} \int_{\gamma} h(\gamma(s), t) ds + O(\varepsilon^2),$$

where γ is the geodesic in the flat metric. As the perturbation decays, $g(x, t) \rightarrow g_0$ smoothly and the arena returns to flatness.

The Agents

Let $\{X_i(t)\}_{i=1}^n$ be a collection of stochastic processes on $\mathbb{T}^2$. Each agent satisfies a stochastic differential equation of the form:

$$dX_i^k(t) = V_i^k(X_i(t), X_j(t - \tau)) dt + \sigma_\ell^k(X_i(t), t) dW_i^\ell(t),$$

where:

- V_i is the **pursuit drift vector field**, steering agent i toward another agent’s position at the delayed time $t - \tau$;
- $\tau > 0$ is the **information delay**—the pursuer reacts to where the prey *was*, not where it *is*;
- $\sigma\sigma^T = g^{-1}(t)$ encodes the geometry of the arena into the noise covariance, so that Brownian fluctuations are isotropic *in the metric sense* at each moment;
- W_i are independent standard Wiener processes.

The generator of this process is the time-dependent operator:

$$\mathcal{L}_t = \frac{1}{2}\Delta_{g(t)} + V_i \cdot \nabla,$$

where $\Delta_{g(t)}$ is the **Laplace-Beltrami operator** of the instantaneous metric—the object that connects stochastic analysis, differential geometry, and the heat equation in a single symbol.

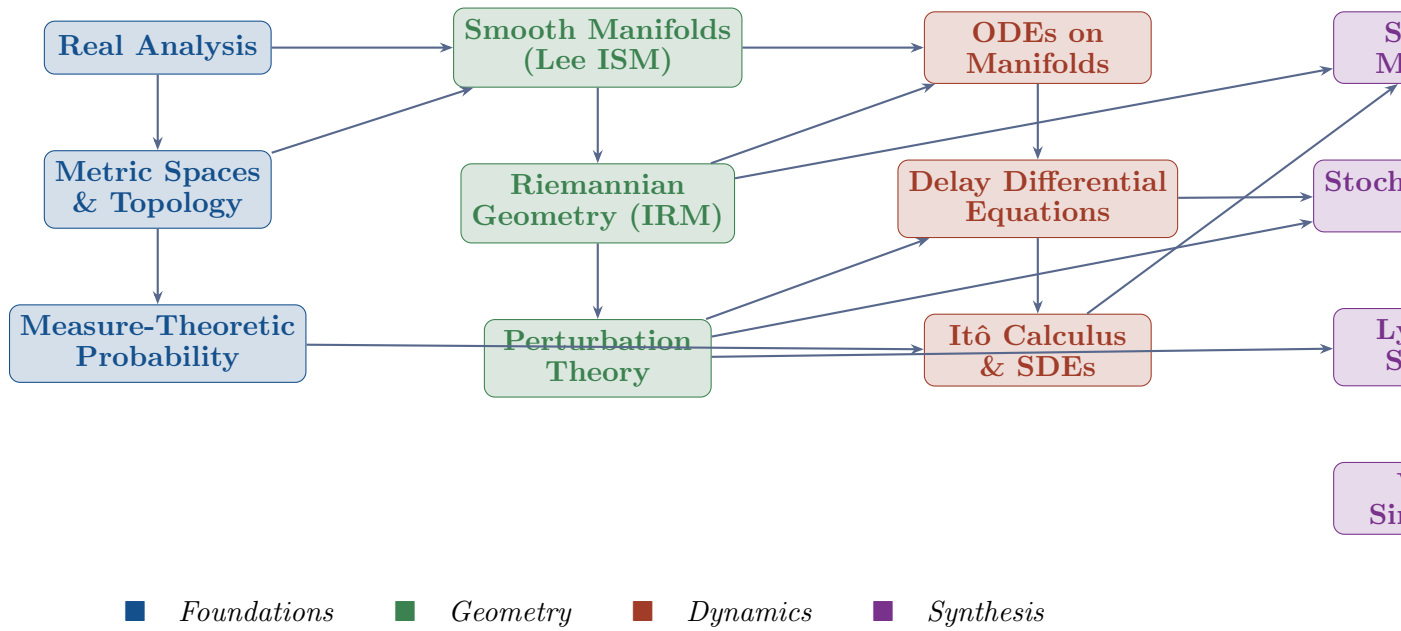
The Central Questions

This project asks, and will eventually answer, the following:

1. **Well-posedness.** Under what conditions on τ , ε , and σ does the system of SDEs admit a unique strong solution on $\mathbb{T}^2$?
2. **Geodesic bifurcation.** When a metric ripple transiently inflates one geodesic between pursuer and prey, does the optimal pursuit path switch to the complementary geodesic? What is the measure of such events?
3. **Stability under perturbation.** In the Lyapunov sense, does the pursuit system remain stable—in expectation and in probability—under the simultaneous action of noise, delay, and metric deformation?
4. **Computational faithfulness.** Can the Vulkan pipeline evaluate the perturbed geodesic distance and the corrected SDE drift at interactive frame rates, with a timestep Δt guaranteed by the Lipschitz constant of the metric field?

Dependency Graph

The following diagram encodes the strict logical dependencies of this project. No node may be treated as understood until all nodes feeding into it are mastered. Arrows denote “is required by.”



Remark 1. Perturbation theory (Stage 2) is not a terminal node. It is a *vertical thread* that re-enters at every subsequent stage: as linearized curvature in Riemannian geometry, as regular and singular perturbation of ODEs and DDEs, as the ε -expansion of the stochastic heat kernel, and as the robustness theory for Lyapunov functions. Its placement in the graph reflects where it is first formally introduced, not where it last appears.

This document is the front matter of an ongoing solo research repository. It will be revised as understanding deepens. The mathematics does not wait for permission to be beautiful.